COLUMBIA UNIVERSITY

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Statistical Inference

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Assignment 7

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Three Bayes Estimators (Problems 1–3)

For each of the following, find the Bayes estimator of the parameter of interest under squared error loss with a sample.

One theorem does the work in all three problems: under squared-error loss, the Bayes estimator is the posterior mean (notes, Thm. 2.9.3; DeGroot and Schervish, Cor. 7.4.1, p. 410; HPS, Thm. 2, p. 38). Each solution therefore has two steps. First find the posterior distribution of the parameter given the data, by matching kernels: the posterior density is proportional to prior times likelihood, and the normalizer never needs computing because the θ-dependent factor already names a standard family (notes, Def. 2.9.1 and Remark 2.9.2). Then take the posterior's mean. Throughout, $Y_1, \ldots, Y_n$ are the data, $s = \sum_i y_i$, and $\bar{y} = s/n$; densities for the Beta and Gamma families follow the Wikipedia pages the assignment points to, with one caution recorded in Problem 3 about the Gamma's two parameterizations.

Problem 1

The data are a sample of n $N(\theta, \sigma^2)$ distribution, θ the parameter of interest, and σ^2 the variance. The prior is that θ has a normal distribution with expectation η and variance τ^2.

Solution:

The prior density is proportional to $e^{-(\theta-\eta)^2/(2\tau^2)}$ and the likelihood, keeping only factors that involve θ, is $\exp[-\sum_i (y_i-\theta)^2/(2\sigma^2)]$. Multiply and collect the exponent as a quadratic in θ:

$$\begin{aligned}
\pi(\theta \mid y) &\propto \exp\left[-\frac{(\theta-\eta)^2}{2\tau^2}\right]\exp\left[-\frac{\sum_i (y_i-\theta)^2}{2\sigma^2}\right] && \text{(prior times likelihood)}\\
&\propto \exp\left[-\frac{\theta^2}{2}\Big(\frac{1}{\tau^2}+\frac{n}{\sigma^2}\Big)+\theta\Big(\frac{\eta}{\tau^2}+\frac{s}{\sigma^2}\Big)\right] && \text{(drop θ-free factors)}\\
&\propto \exp\left[-\frac{1}{2v}\big(\theta-m\big)^2\right], \qquad \frac{1}{v}=\frac{1}{\tau^2}+\frac{n}{\sigma^2}, \quad m=v\Big(\frac{\eta}{\tau^2}+\frac{n\bar{y}}{\sigma^2}\Big) && \\
& && \text{(complete the square)}
\end{aligned}$$

This is the $N(m,v)$ kernel, so the posterior is normal and its mean is m. Clearing v,

$$m \;=\; \frac{\eta/\tau^2+n\bar{y}/\sigma^2}{1/\tau^2+n/\sigma^2} \;=\; \frac{\sigma^2\,\eta+n\tau^2\,\bar{y}}{\sigma^2+n\tau^2},$$

and by the posterior-mean theorem the Bayes estimator is

$$\boxed{\delta(Y) \;=\; \frac{\sigma^2\,\eta \;+\; n\tau^2\,\bar{Y}}{\sigma^2+n\tau^2} \;=\; \frac{n\tau^2}{n\tau^2+\sigma^2}\,\bar{Y} \;+\; \frac{\sigma^2}{n\tau^2+\sigma^2}\,\eta}$$

a weighted average of the sample mean and the prior expectation. The weights are the two precisions: the data carry n/σ^2 (the Fisher information of Assignment 5) and the prior carries $1/\tau^2$, and each side of the average gets its share. A vague prior ($\tau^2 \to \infty$) hands everything to $\bar{Y}$; a dogmatic one ($\tau^2 \to 0$) ignores the data. Assignment 3's normal problem was the case $\eta = 1$, $\tau^2 = 1$, where this estimator appeared as the best member of the shrinkage family. The single-observation version, with the full completing-the-square, is Example 2.9.4 and Appendix 2.A.4 of the notes; DeGroot and Schervish state the n-sample update as Theorem 7.3.3 (pp. 398–400), and HPS works it in §2.8 (pp. 39–41), including the same weighted-average reading. ■

Problem 2

The data are a sample of n Bernoulli(p) variables, where p is the unknown parameter of interest, and we suppose that p has a Beta distribution with parameters α and β.

SOLUTION:

The Beta(α, β) prior has density proportional to $p^{\alpha-1}(1-p)^{\beta-1}$ on $(0,1)$, with mean $\alpha/(\alpha+\beta)$. The likelihood of a Bernoulli sample with s successes is $p^s(1-p)^{n-s}$. The two kernels multiply into a third:

$$\begin{aligned} \pi(p \mid y) &\propto p^{\alpha-1}(1-p)^{\beta-1} \cdot p^s(1-p)^{n-s} && \text{(prior times likelihood)} \\ &= p^{(\alpha+s)-1}\,(1-p)^{(\beta+n-s)-1} && \text{(add exponents)} \end{aligned}$$

which is the Beta$(\alpha+s,\ \beta+n-s)$ kernel: successes are added to α, failures to β. The posterior mean is $(\alpha+s)/(\alpha+\beta+n)$, so

$$\boxed{\delta(Y) \;=\; \frac{\alpha+S}{\alpha+\beta+n} \;=\; \frac{n}{\alpha+\beta+n}\,\bar{Y} \;+\; \frac{\alpha+\beta}{\alpha+\beta+n}\cdot\frac{\alpha}{\alpha+\beta}}$$

again a weighted average of the sample rate and the prior mean, with the prior counting for $\alpha+\beta$ pseudo-observations against the data's n. The uniform prior is $\alpha=\beta=1$, and the estimator becomes $(1+S)/(2+n)$: the shaded estimator this course has been carrying since Assignment 1, and Laplace's rule of succession (Pitman, pp. 420–421). Figure 1 shows one update. The result matches Example 2.9.5 of the notes, Theorem 7.3.1 and Example 7.4.3 of DeGroot and Schervish (pp. 394–395, 410), Equation (27) of HPS (p. 39), and the conjugacy story of Blitzstein and Hwang (Story 8.3.3, pp. 383–385).

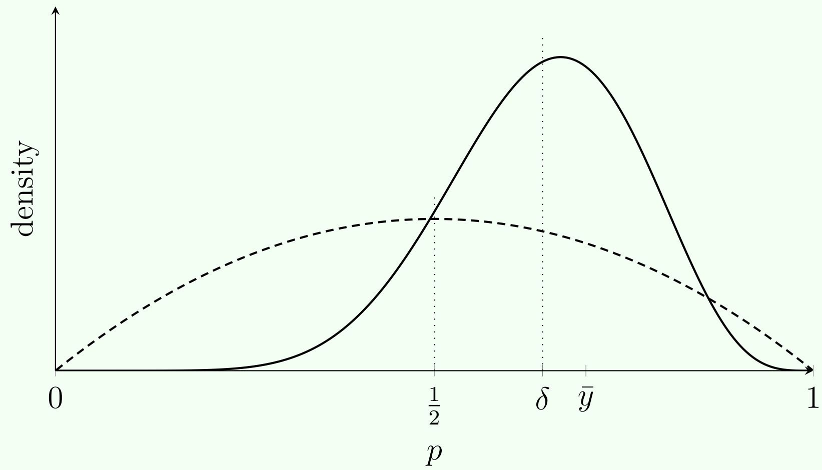

Figure 1. ONE CONJUGATE UPDATE. A Beta$(2, 2)$ prior (dashed, with mean $\frac{1}{2}$ marked) meets $n = 10$ Bernoulli trials with $s = 7$ successes and becomes the Beta$(9, 5)$ posterior (solid). The Bayes estimate $\delta = \frac{9}{14} \approx 0.643$ sits between the prior mean and the sample rate $\bar{y} = 0.7$, closer to the data because n outweighs $\alpha + \beta = 4$.

$\blacksquare$

Problem 3

The data are a sample from a Poisson distribution with parameter λ, and the prior on λ is Gamma with parameters α and β.

SOLUTION:

The Wikipedia page the assignment points to gives the Gamma distribution in two parameterizations, shape–scale (k, θ) and shape–rate (α, β). Since the assignment names the parameters α and β, the shape–rate convention applies:

$$\pi(\lambda) \;=\; \frac{\beta^\alpha}{\Gamma(\alpha)}\,\lambda^{\alpha-1}e^{-\beta\lambda}, \qquad \lambda > 0,$$

with mean α/β. (In the scale convention the same prior would be written Gamma(α, $1/\beta$); the posterior below converts accordingly.)

The likelihood of a Poisson sample, keeping only the λ-dependent factors, is $e^{-n\lambda}\lambda^s$. Matching kernels once more:

$$\begin{aligned} \pi(\lambda \mid y) \;&\propto\; \lambda^{\alpha-1}e^{-\beta\lambda}\cdot e^{-n\lambda}\lambda^s && \text{(prior times likelihood)}\\ &=\; \lambda^{(\alpha+s)-1}\,e^{-(\beta+n)\lambda} && \text{(add exponents)} \end{aligned}$$

the Gamma($\alpha + s$, $\beta + n$) kernel: the total count joins the shape, the sample size joins the rate. The posterior mean is the shape over the rate, so

$$\boxed{\delta(Y) \;=\; \frac{\alpha+S}{\beta+n} \;=\; \frac{n}{\beta+n}\,\bar{Y} \;+\; \frac{\beta}{\beta+n}\cdot\frac{\alpha}{\beta}}$$

one more weighted average: sample mean against prior mean α/β, with the prior worth β observations' exposure and α events. This is Example 2.9.6 of the notes.

The three problems come out identically because the same two facts drive them. Squared-error loss makes the Bayes estimator a posterior mean, and each prior is conjugate to its model, so the posterior stays in the prior's family and its mean is a closed-form compromise

$$\delta \;=\; w\,\bar{Y} + (1-w)\times(\text{prior mean}), \qquad w = \frac{n}{n + \text{prior weight}},$$

where the prior's weight is σ^2/τ^2 equivalent observations in Problem 1, $\alpha+\beta$ pseudo-trials in Problem 2, and β units of exposure in Problem 3. In every case $w \to 1$ as n grows: the data buy out the prior at rate $1/n$, which is the same arithmetic that made Assignment 3's optimal shrinkage weight $\sigma^2/(n+\sigma^2)$ vanish in the limit. ■

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